

# Performing a Vulnerability Assessment (4e)

Fundamentals of Information Systems Security, Fourth Edition - Lab 02

Student:

Alexander Cher-Aime

Email:

acheraime@usf.edu

Time on Task:

6 hours, 8 minutes

Progress:

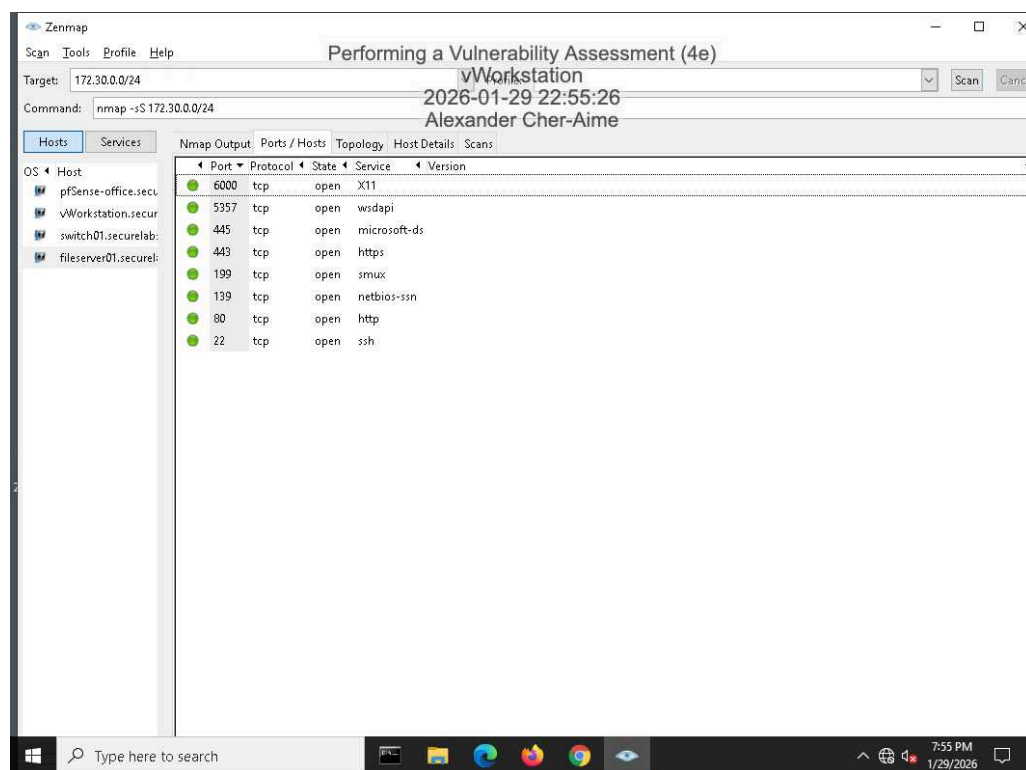
100%

Report Generated: Sunday, February 1, 2026 at 10:37 PM

## Section 1: Hands-On Demonstration

### Part 1: Scan the Network with Zenmap

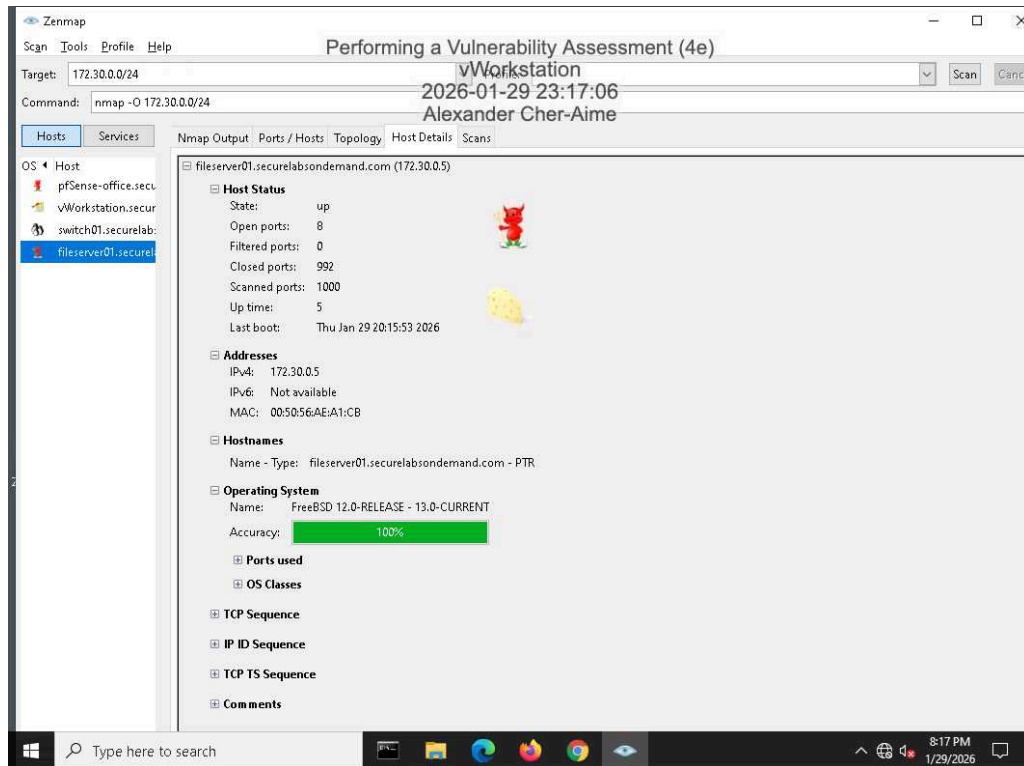
9. Make a screen capture showing the contents of the **Ports/Hosts** tab from the **SYN** scan for **fileserver01.securelabsondemand.com**.



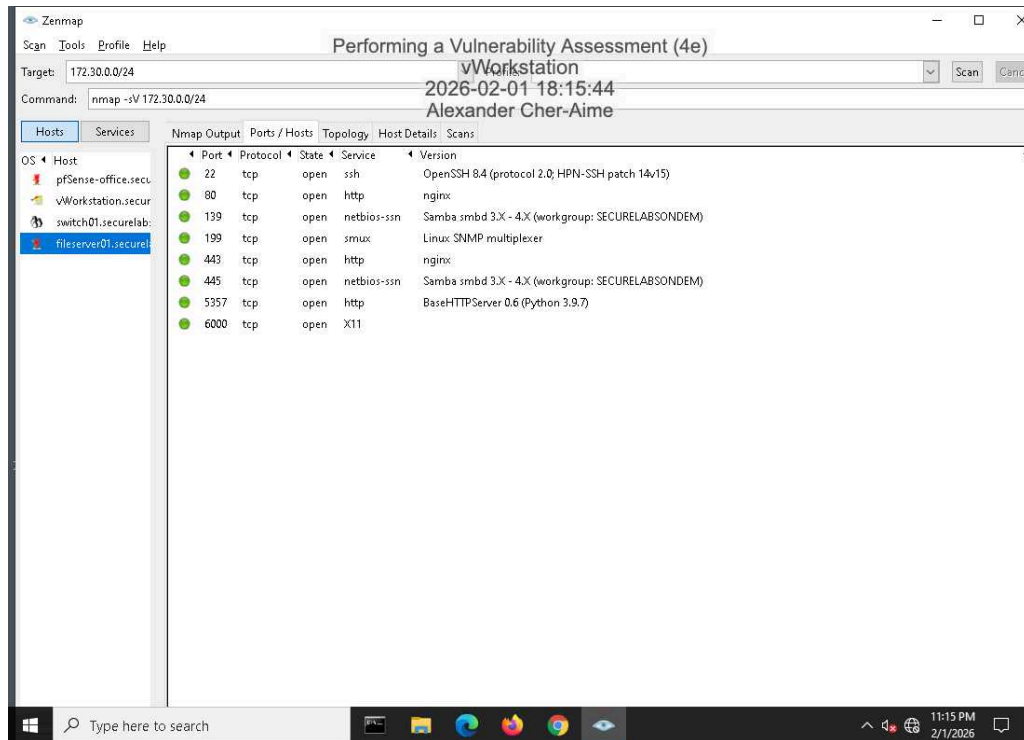
# Performing a Vulnerability Assessment (4e)

Fundamentals of Information Systems Security, Fourth Edition - Lab 02

15. Make a screen capture showing the contents of the **Host Details** tab from the OS scan for **fileserver01.securelabsondemand.com**.

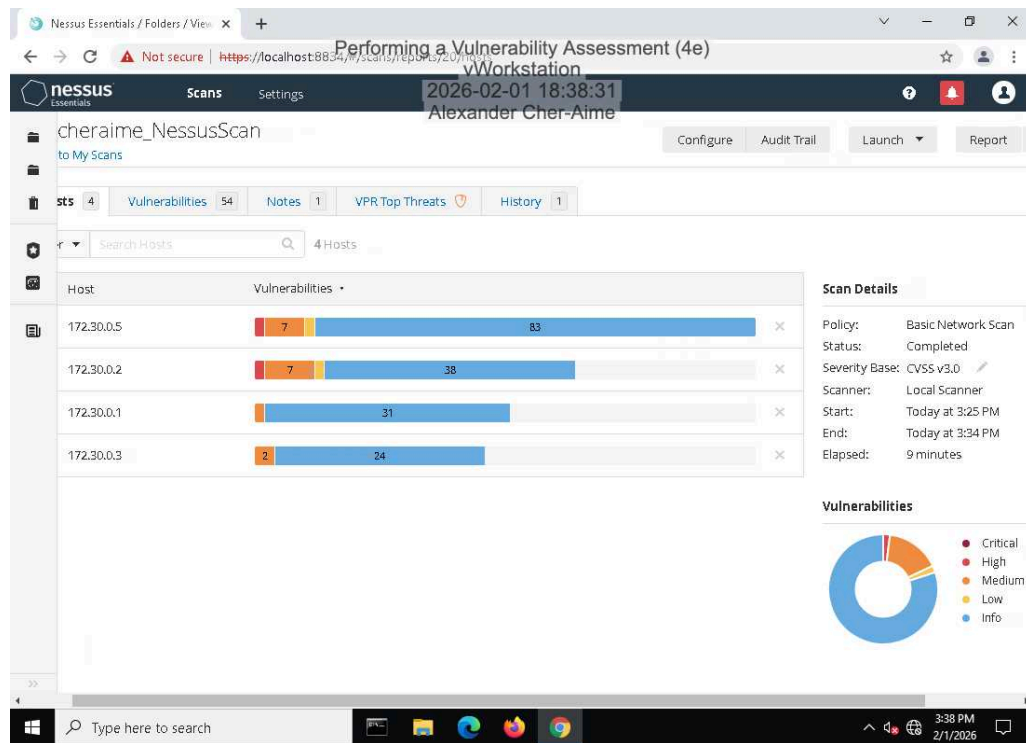


19. Make a screen capture showing the details in the **Ports/Hosts** tab from the **Service scan** for **fileserver01.securelabsondemand.com**.



## Part 2: Conduct a Vulnerability Scan with Nessus

### 14. Make a screen capture showing the Nessus report summary.



## Part 3: Evaluate Your Findings

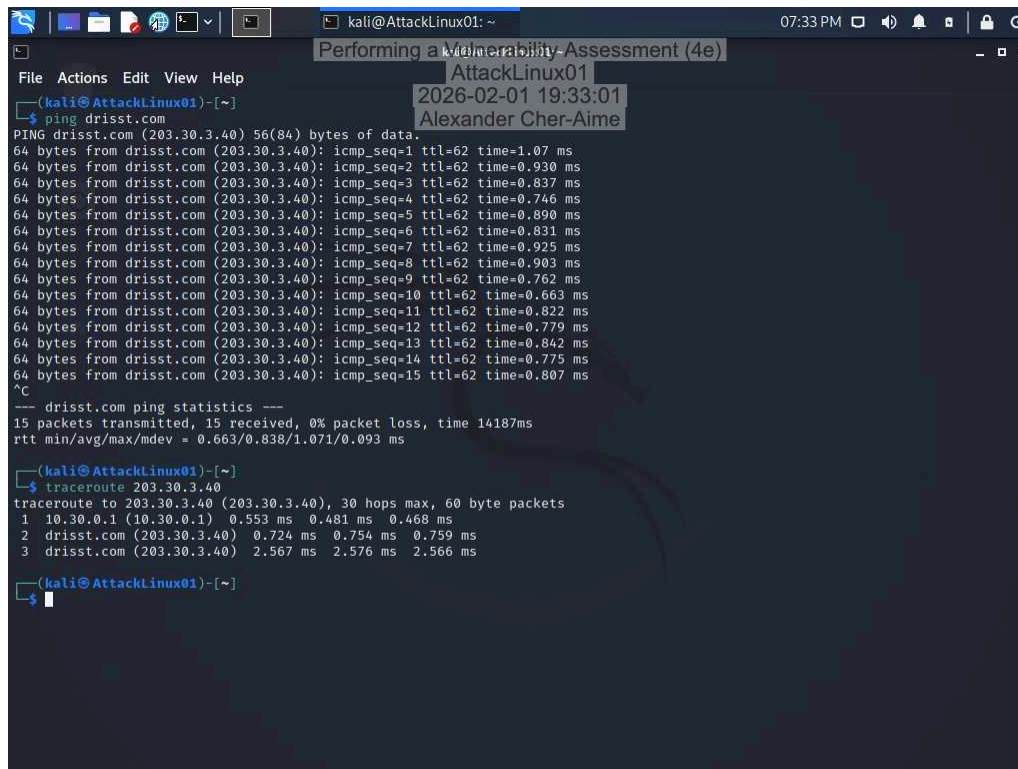
### 11. Summarize the vulnerability you selected, including the CVSS risk score, and recommend a mitigation strategy.

The vulnerability I chose was the plugin 45411 SSL Certificate with Wrong Hostname. The CVSS risk score is 5.3, the recommend strategy is to purchase or generate a proper SSL certificate for this service.

### Section 2: Applied Learning

#### Part 1: Scan the Network with Nmap

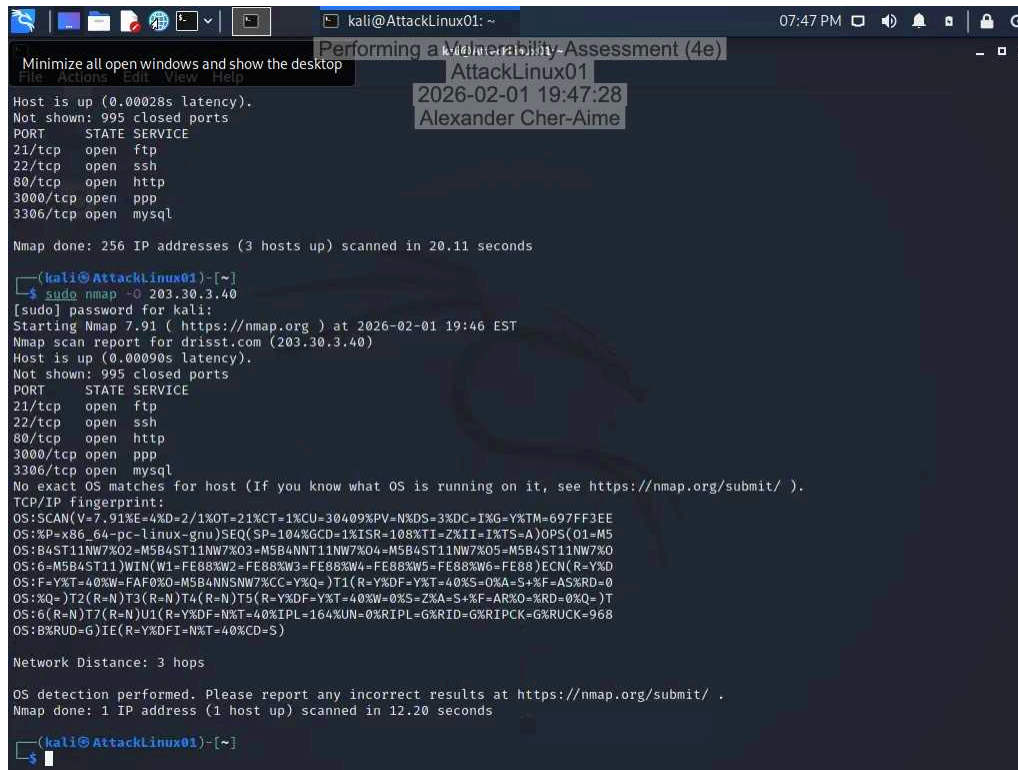
6. Make a screen capture showing the results of the traceroute command.



The screenshot shows a Kali Linux terminal window with the following content:

```
kali@AttackLinux01: ~  
Performing a Vulnerability Assessment (4e)  
AttackLinux01  
2026-02-01 19:33:01  
Alexander Cher-Aime  
File Actions Edit View Help  
kali@AttackLinux01: ~  
$ ping drisst.com  
PING drisst.com (203.30.3.40) 56(84) bytes of data:  
64 bytes from drisst.com (203.30.3.40): icmp_seq=1 ttl=62 time=1.07 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=2 ttl=62 time=0.930 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=3 ttl=62 time=0.837 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=4 ttl=62 time=0.746 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=5 ttl=62 time=0.890 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=6 ttl=62 time=0.831 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=7 ttl=62 time=0.925 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=8 ttl=62 time=0.903 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=9 ttl=62 time=0.762 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=10 ttl=62 time=0.663 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=11 ttl=62 time=0.822 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=12 ttl=62 time=0.779 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=13 ttl=62 time=0.842 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=14 ttl=62 time=0.775 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=15 ttl=62 time=0.807 ms  
^C  
--- drisst.com ping statistics ---  
15 packets transmitted, 15 received, 0% packet loss, time 14187ms  
rtt min/avg/max/mdev = 0.663/0.838/1.071/0.093 ms  
kali@AttackLinux01: ~  
$ traceroute 203.30.3.40  
traceroute to 203.30.3.40 (203.30.3.40), 30 hops max, 60 byte packets  
1 10.30.0.1 (10.30.0.1) 0.553 ms 0.481 ms 0.468 ms  
2 drisst.com (203.30.3.40) 0.724 ms 0.754 ms 0.759 ms  
3 drisst.com (203.30.3.40) 2.567 ms 2.576 ms 2.566 ms  
kali@AttackLinux01: ~  
$
```

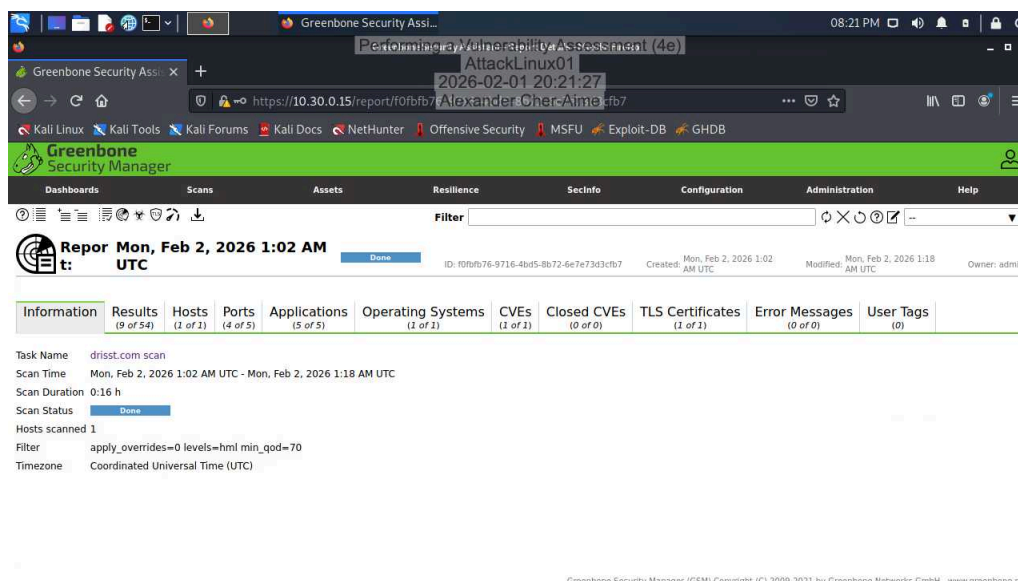
10. Make a screen capture showing the results of the Nmap scan with OS detection activated.



```
kali@AttackLinux01: ~  
Performing a Vulnerability Assessment (4e)  
AttackLinux01  
2026-02-01 19:47:28  
Alexander Cher-Aime  
Minimize all open windows and show the desktop  
File Actions Edit View Help  
Host is up (0.00028s latency).  
Not shown: 995 closed ports  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
80/tcp    open  http  
3000/tcp  open  ppp  
3306/tcp  open  mysql  
  
Nmap done: 256 IP addresses (3 hosts up) scanned in 20.11 seconds  
  
kali@AttackLinux01: ~  
[sudo] password for kali:  
Starting Nmap 7.91 ( https://nmap.org ) at 2026-02-01 19:46 EST  
Nmap scan report for drisst.com (203.30.3.40)  
Host is up (0.00090s latency).  
Not shown: 995 closed ports  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
80/tcp    open  http  
3000/tcp  open  ppp  
3306/tcp  open  mysql  
  
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).  
TCP/IP fingerprint:  
OS:SCAN(V-7.91%E-4%Q-2/1%OT-21%CT-1%CU-304.09%PV-WKDS-3%DC-1%G-Y%TM-697FF3EE  
OS:XP-x86_64-vc-linux-gnu)SEQ(SP-104%GCD-1%ISR-108%TI-2%T1-1%TS-A)OP5(O1-H5  
OS:84ST11NW7%O2-M5B4ST11NW7%O3-M5B4NW11NW7%O4-M5B4ST11NW7%O5-M5B4ST11NW7%O  
OS:6-M5B4ST11WIN(W1-FE88%W2-FE88%W3-FE88%W4-FE88%W5-FE88%W6-FE88%W7-ECN(R-Y%  
OS:6-Y%T-10%W-FAF0%O-M5B4NWSNW7%CC-Y%Q-)T1(R-Y%DF-Y%T-40%S-0%A-S+%F-AS%RD-0  
OS:9Q-)T2(R-N)T3(R-N)T4(R-N)T5(R-Y%DF-Y%T-40%W-0%S-Z%A-S+%F-AR%O-%RD-0%Q-)T  
OS:6(R-N)T7(R-N)UI(R-Y%DF-N%T-40%IPL-164%UN-0%RIPL=G%RID=G%RIPCK=G%RUCK-968  
OS:8%RUD=G)IE(R-Y%DFI-N%T-40%CD-S)  
  
Network Distance: 3 hops  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 12.20 seconds  
  
kali@AttackLinux01: ~  
[sudo]
```

## Part 2: Conduct a Vulnerability Scan with OpenVAS

13. Make a screen capture showing the detailed OpenVAS scan results.



## Part 3: Prepare a Penetration Test Report

### Target

Insert the target here.

MySQL / MariaDB weak password, Anonymous FTP Login Reporting, and Missing `httpOnly` Cookie Attribute

### Completed by

Insert your name here.

Alexander Cher-Aime

### On

Insert current date here.

2/1/26

### Purpose

Identify the purpose of the penetration test.

to identify the highest vulnerabilities

### Scope

Identify the scope of the penetration test.

The scope of this is to summarize the findings on the three highest severity vulnerability.

### Summary of Findings

Identify and summarize each of the three high-severity vulnerabilities identified during your penetration test. For each vulnerability, identify the severity, describe the issue, and recommend a remediation.

First one the severity level is 9.0(95%). It was due to a weak password, it was possible tp login as root with password "password". The best solution is to change the password as soon as possible. Second one has a severity of 6.4(80%). Reports if the remote FTP Server allows anonymous logins. the issue was that it was possible to login with the anonymous accounts. The solution would be to disable anonymous logins. The third one is missing the 'httpOnly' cookie attribute. It's severity is 5(80%), the issue is due to the httpOnly attribute missing. The cookie can be accessed by JavaScript which can lead to being attacked. The solution is to set the httpOnly attribute for any session cookie.

### Conclusion

Identify your key findings.

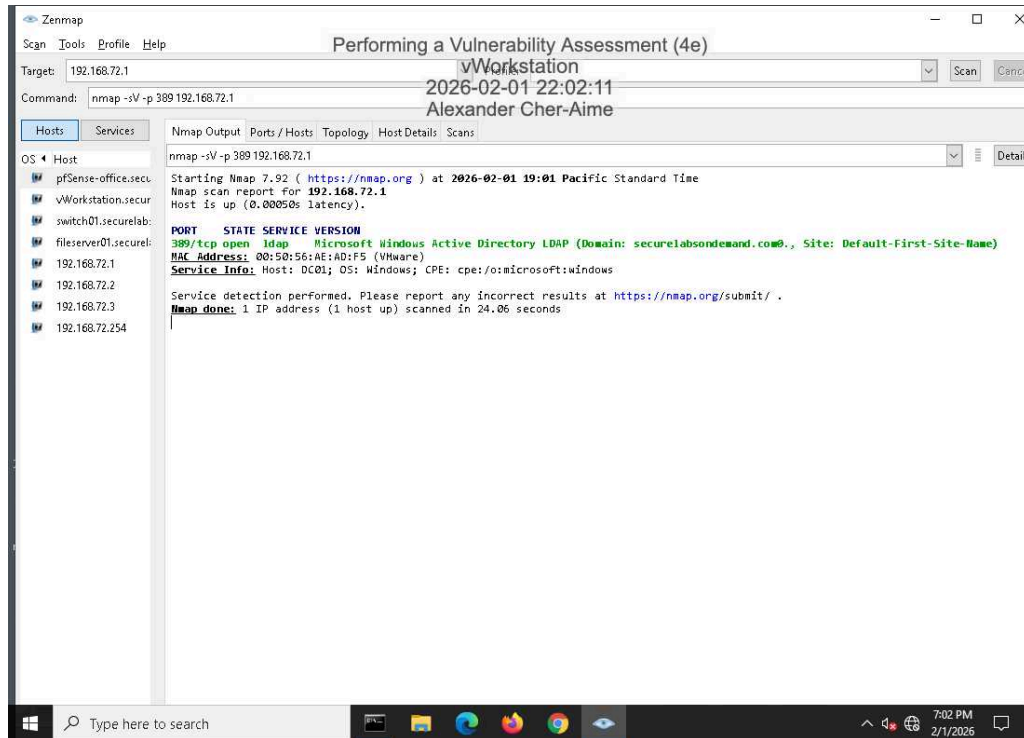
My key findings that all of the vulnerabilities have solutions to them, so you can always find a way to fix them. Just having you password as "password" can be consider a high severity. Each vulnerability has a detection method listed as well. They are also in order by severity.



### Section 3: Challenge and Analysis

#### Part 1: Scan the Domain Controller with Nmap

Make screen capture showing the results of your targeted port scan on the domain controller.

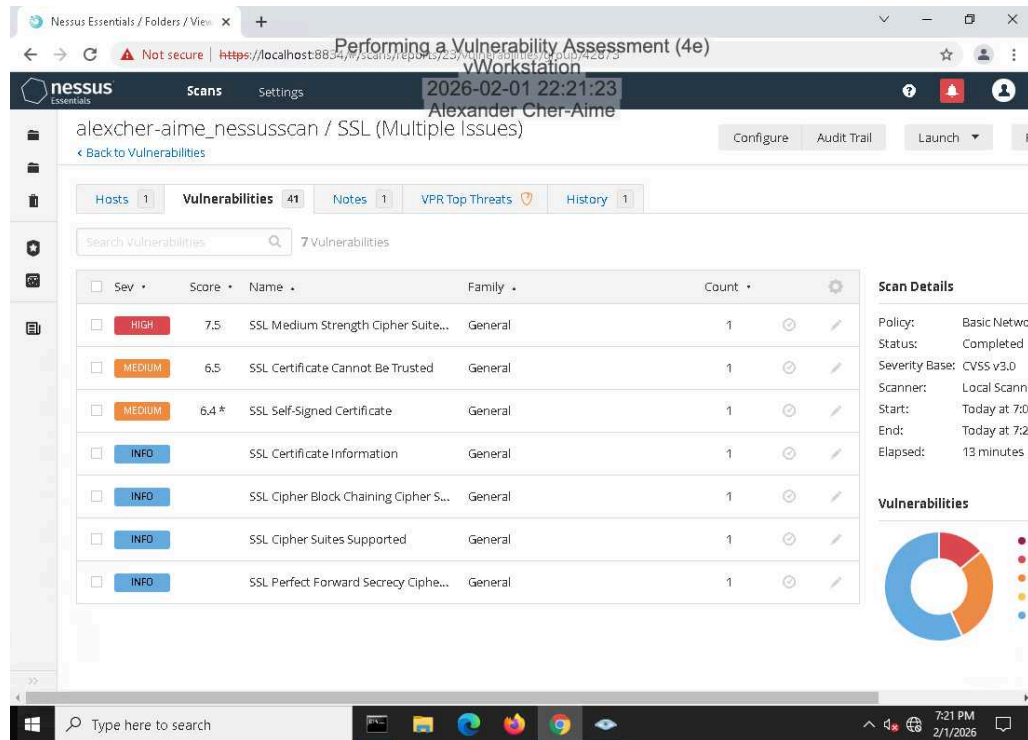


#### Part 2: Scan the Domain Controller with Nessus

# Performing a Vulnerability Assessment (4e)

Fundamentals of Information Systems Security, Fourth Edition - Lab 02

Make a screen capture showing the Nessus report summary for the domain controller.



## Part 3: Prepare a Penetration Test Report

### Target

Insert the target here.

One high severity vulnerability, SSL Medium Strength Cipher Suites Supported (SWEET32)

### Completed by

Insert your name here.

Alexander Cher-Aime

### On

Insert current date here.

02/01/2026

### Purpose

Identify the purpose of the penetration test.

Purpose of this test is to identify the high severity.

### Scope

Identify the scope of the penetration test.

The scope is to analyze the vulnerability SSL Medium Strength Cipher Suites Supported (SWEET32), and find a solution.

### Summary of Findings

Identify and summarize each vulnerability identified during your penetration test. For each vulnerability, identify the severity, describe the issue, and recommend a remediation.

SSL Medium Strength Cipher Suites Supported (SWEET32) is vulnerability with a high severity, CVSS base score of 7.5. The remote host supports the use of SSL ciphers that offer medium strength encryption. The issue is that it's considerably easier to circumvent medium strength encryption if the attacker is on the same physical network. The solution to this is to reconfigure the affected application if possible to avoid use of medium strength ciphers.

### Conclusion

Identify your key findings.

My key findings were that this vulnerability was mixed with other ones. If not checked properly this could have gone unnoticed and caused an attack. Nessus regards medium strength as any encryption that uses key lengths at least 64 bits and less than 112 bits, or else that uses the 3DES encryption suite. This plugin is also listed as outdated.